



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **EEE F411**
Course Title : **Internet of Things**
Instructor-in-Charge : **Prof. Vinay Chamola**
Instructor(s) : **none**
Tutorial/Practical Instructors: **Manvendra Singh, Shubham Sharma and Manish prasad**

- 1 Course Description:** Introduction to IoT, Sensors and Actuators for IoT, Programming IoT end points, Communications and networking in IoT, Data management in IoT, Security issues in IoT, Emerging technologies. This course is **discipline elective (DEL)** for: **Undergraduate level:** B.E. (EEE), B.E (ECE), B.E. (E&I)/ dualites in those disciplines. **For other discipline, it will be considered as an open elective (OPEL).**

2 Scope and Objective of the Course:

This module is designed to provide students with solid technical knowledge and skills to build Internet of Things (IoT) systems. Internet of things has evolved due to convergence of multiple technologies - embedded systems, sensor technology, real-time data analytics, machine learning etc. Traditional fields of embedded systems, wireless sensor networks, control systems, automation (including home and building automation), and others all contribute to enabling the IoT. This course comprehensively covers various technologies and tools used for enabling IoT solutions. Knowledge of various topics required for building IoT prototypes like sensors and actuators/ Communications and networking and data management is also imparted in this course. This course would also help the students understand the various IoT security challenges and solution to address them. The course will also give the students exposure to how various real world problems are being solved by IoT based solutions (like in applications for smart city, smart farming etc.). There would also be some hands on sessions where students would learn how to build and program IoT systems and make end-to- end solutions for different applications. Furthermore, assignments and projects in this course would help students build IoT prototypes and apply what they have learnt in the course to solve real world problems.

3 Course learning Objectives: Upon completion of this course, students will be able to:

- Understand and explain** the fundamental concepts, architectures, enabling technologies, communication protocols, data-management techniques, and security challenges associated with Internet of Things systems.
- Design and develop** IoT-based solutions by integrating sensors, actuators, microcontrollers, communication networks, cloud platforms, and data-analytics tools.
- Apply practical programming and system-integration skills** to build, test, and evaluate end-to-end IoT prototypes for addressing real-world applications such as smart cities, smart agriculture, healthcare, and industrial automation.

4 Text Books:

- (T1) *Internet of Things: Principles and paradigms*. R. Buyya, and A.V Dastjerdi (Elsevier), 2016.
(R1) "Precision - Internet Of Things", by Timothy Chou (Mc Graw Hill), 2017.
(R2) "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press), 2017.
(R3) "Internet of Things: A Hands-on Approach", by Arshdeep Bahga and Vijay Madisetti (Universities Press), 2014.
(R4) "Internet of Things" by Raj Kamal (Mc Graw Hill), 2017.



5. Course Plan:

Module No.	Lecture Session	References	Topics covered/ Learning Outcomes
Module 1: Introduction to IoT	Lectures 1- 4	Section 1.2(T1), Chapter 1,2 (R3), Section 5.2, 9.3 (R3), Chapter 1 (R4), relevant online resources	Introduction to IoT & Cyber-Physical Systems, IoT applications - an overview, Different Levels of IoT Applications: Level 1 - 6 with examples, IoT Design Methodology & Life Cycle, Introduction to IoT Physical End Points & Platforms, IoT System Design Examples (for applications like fitness tracker, smart parking etc.). Learning these concepts to understand how end to end IoT applications work.
Module 2: Sensors and Actuators for IoT	Lectures 5 - 8	Section 1.7, 15.4(T1), Chapter 7 (R4), relevant online resources	Introduction to sensors for IoT application development, Data Acquisition, Signal Conditioning and Processing, Multi Sensor fusion for IoT, Advanced sensing techniques (e.g. BCI/HCI), Actuators and Controllers for IoT. Understand their working and how to apply them to develop novel technology solutions
Module 3: Programming IoT end points	Lectures 8- 18	Chapter 6,7 (R3), Appendix A (R3), relevant online resources	Introduction to Arduino microcontroller & NodeMCU for IoT applications, Programming with Arduino & NodeMCU and prototype development (e.g. for smart farming, smart city applications etc.), Introduction to Raspberry Pi, Programming Raspberry Pi (Python), Rest API based hands on. Getting a hands on experience in these tools to enhance skill set.
Module 4: Networking for IoT	Lectures 18 - 23	Section 1.4.1 (R3), Chapter 7(R4), relevant online resources	Understanding wireless sensor networks: Introduction, Deployment, Localization, Routing, MAC, Time synchronization and Multi-sensor-fusion for WSN. Blockchain fundamentals and it's applications in IoT
Module 5: Communications and networking in IoT	Lectures 24-31	Chapter 16(T1) , Section 1.2.3 (T1), Section 12.4 (R4), relevant online resources	Get an overview of Ad Hoc networks (MANET, VANET, FANET), Industrial IoT Networks, Common network standards (Bluetooth, NFC, LORA) etc, COAP and MQTT (including lab/ hands-on)
Module 6: Data management in IoT	Lectures 32-36	Section 1.5 (T1) , Chapter 5,6(R4), relevant online resources	Understanding data management for IoT, Advanced optimization for processing sensor data, Machine learning for IoT data analytics, Introduction to IoT Cloud Services, Case studies of Cloud services for IoT and learning how to use them.
Module 7: Security issues in IoT	Lectures 37 - 40	Chapter 10, Section 1.8(T1), Chapter 10 (R4), relevant online resources	Get an overview of cyber-attacks on IoT- Case study and understand security solutions for IoT: hardware/software



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Lab:

Lab for the course would be in-class labs during the class hours when the relevant topics are covered. Labs would consist of hands on session where the students will learn how to work with microcontrollers and raspberry pi and learn building IoT applications from the scratch. Some of the suggested lab exercises are:

- | Programming Arduino/ Nodemcu (esp8266) microcontrollers.
- | Blinking LED using the above microcontrollers.
- | Interfacing sensors to microcontrollers.
- | Connecting microcontrollers to the internet and streaming sensor data to cloud.
- | Actuation using microcontrollers.
- | Raspberry pi programming and application development.
- | Rest API based hands on.
- | Lab exercise based on COAP/ MQTT.
- | IoT Security based labs

6 Evaluation Scheme: (*minor changes may be done in weightage of components during the semester)

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	1.5 hour	30	Please refer timetable	CB
Lab Test	1 hour	15	TBA	OB
Project (Prototype implementation)	Will be announced	15	Continuous evaluation	OB
Comprehensive Examination	2 hour	40	Please refer timetable	CB

7. Chamber Consultation Hour: To be announced in Class

8 Notices: All notices regarding the course will be mailed.

9. Make-up Policy: No make-up without prior permission

10. Note (if any): This course is **discipline elective** for: **Undergraduate level** : B.E. (EEE), B.E. (E&I). **For other discipline, it will be considered as an open elective.**

Instructor-in-charge
Course No. EEE F411